

Chapter 7

Circles, Arcs, and Sectors

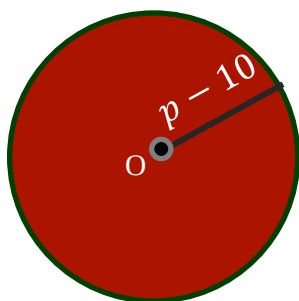


Figures are not drawn to scale. If units are not specified, measurements are in units.

1) How many degrees are eight and a half rotations traveling around a circle?

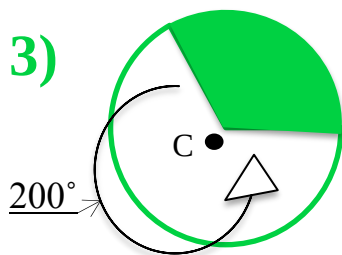
- A) 2080 B) 3060 C) 4080 D) 520 E) 920

2) If the Circumference of Circle O is 30π , what is the value of p ?



- A) 20 B) 15 C) 25 D) 30 E) 35

3) In Circle C, what fraction of the circle is shaded?



- A) $\frac{3}{8}$ B) $\frac{2}{5}$ C) $\frac{2}{7}$ D) $\frac{3}{10}$ E) $\frac{4}{9}$

1	2	3	4	5	6	7	8	9	10
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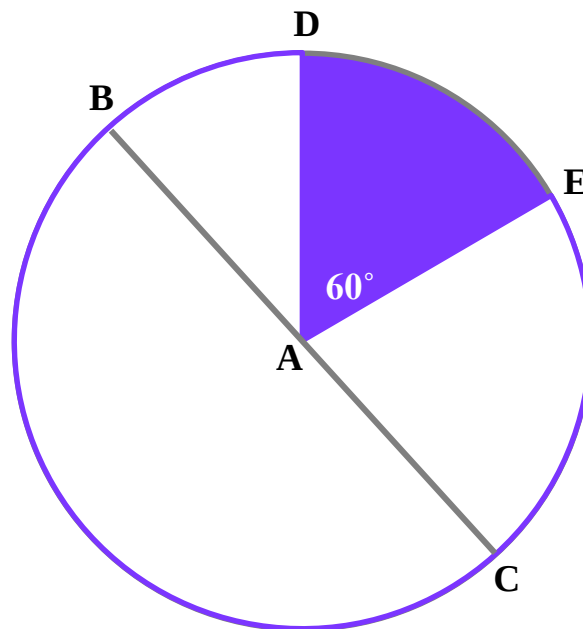
4) If a round birthday cake were cut into 40 equal slices, what must be the central angle (the tip) of each slice?

- A) Between 10° and 15°
- B) Between 22° and 27°
- C) Between 5° and 10°
- D) Greater than 30°
- E) Between 15° and 20°

5) In a standard analog clock, what is the central angle formed from the minute and hour hand when the minute hand is on the four and the hour hand is on the 6?

- A) 60°
- B) 75°
- C) 120°
- D) 150°
- E) 210°

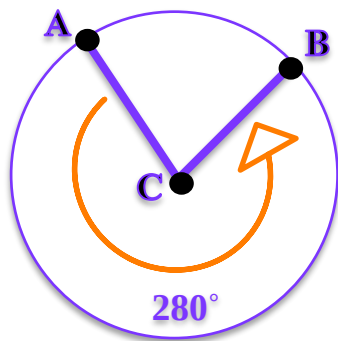
6)



If A is the center of the circle above and $\overline{BC} = 12$, what is the length of Arc $\widehat{DE}$?

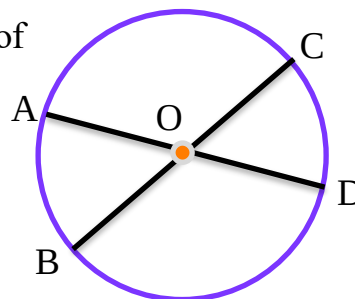
- A) 3π
- B) 2π
- C) $6 - 3\pi$
- D) 12π
- E) $6 - 2\pi$

7) In Circle C, if segment $\overline{AC} = 6\text{mm}$, what is the length of arc $\widehat{AB}$?



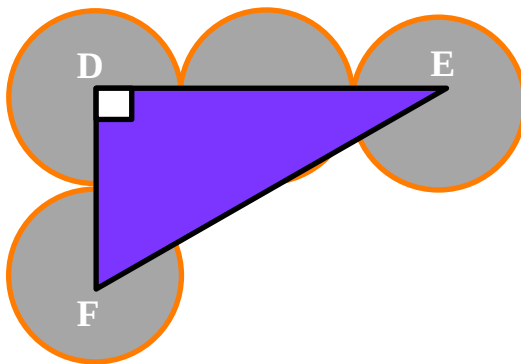
- A) $\frac{8}{3}\pi$ B) 6π C) $\frac{12}{7}\pi$ D) $\frac{7}{3}\pi$ E) $\frac{2}{5}\pi$

8) In the figure to the right, if O is the center of Circle O, segment $\overline{AO} = 8$, and Arc $\widehat{AB} = 45^\circ$, what is the area of sector AOC in squared units?



- A) 24π B) 12π C) $3\pi/8$ D) $5\pi/11$ E) 12

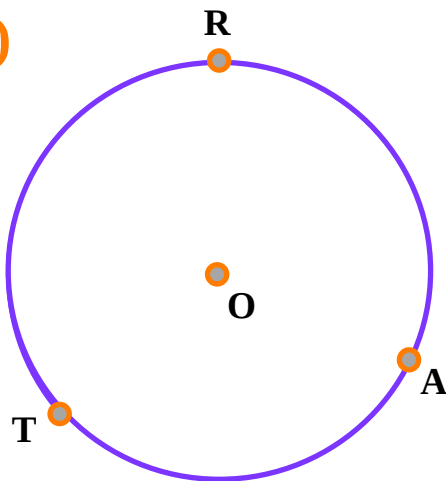
9)



In the previous figure, all circles are tangent with equal area, and each vertex of right triangle DEF is the center of a circle. What is the length of side $\overline{EF}$ if each circle has a circumference of 4π ?

- A) $4\sqrt{5}$ B) 10 C) $10\sqrt{5}$ D) 12 E) $5\sqrt{5}$

10)



In Circle O to the left, if the length of segment $\overline{OR}$ is 9mm, and the measure of angle ROT is 120° , what is the length of Arc RAT?

- A) 4π B) 6π C) 8π D) 10π E) 12π

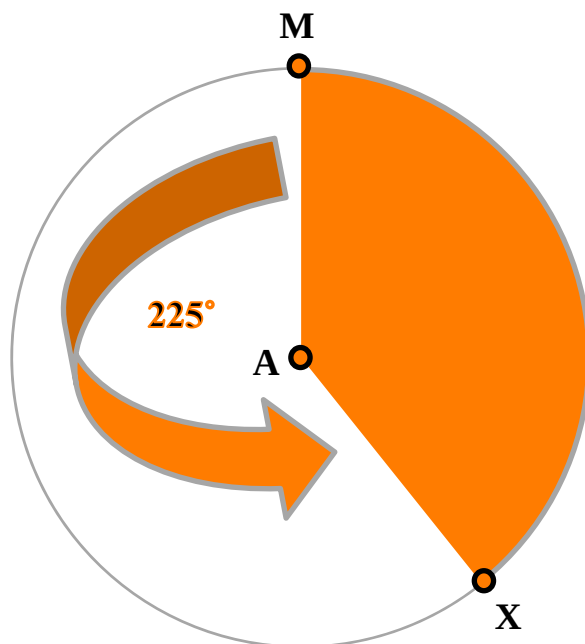
11)



In the previous figure, rectangle ABCD is attached to a semi-circle. If the area of ABCD is 96, and the length of $\overline{AB} = 12$ units, what is the perimeter of the region in orange?

- A) $8 + 6\pi$
 B) 6π
 C) $8 + 4\pi$
 D) $6 + 6\pi$
 E) $8 + 8\pi$

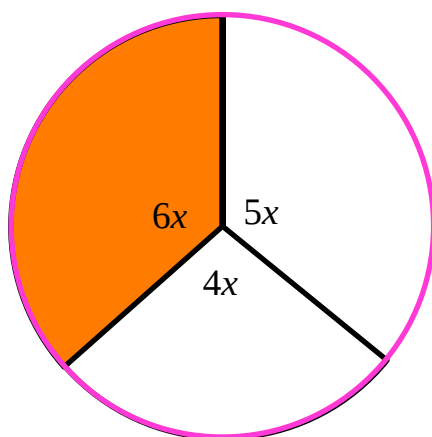
12)



If the Area of Sector MAX is 3π , what is the length of $\overline{AX}$?

- A) 3 B) $2\sqrt{2}$ C) $\sqrt{10}$ D) $4\sqrt{3}$ E) 4

13)

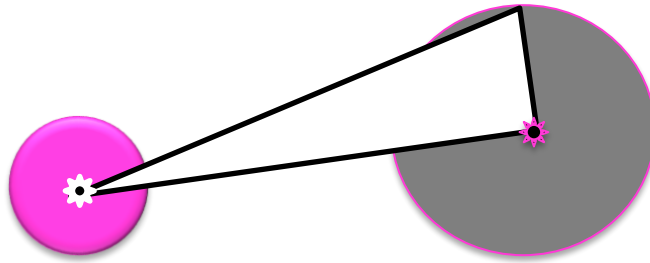


In the circle above, the central angles are listed ($4x$, $5x$, and $6x$). What is the area of the orange region if the radius of the circle is r ?

- A) $20\pi r^2$ B) $\frac{\pi r^2}{6}$ C) $\frac{\pi r^2}{4}$ D) $\frac{\pi r^2}{8}$ E) $\frac{2\pi r^2}{5}$

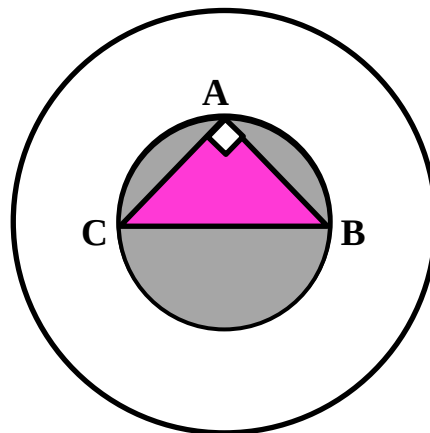
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14) If the area of the gray tire is $\frac{81}{4\pi}$ Feet and its radius is three times the length of the radius of the pink tire, and the geometric contraption(gc) below rolls along the ground without slipping; how many rotations will the pink tire make if the (gc) moves 36 feet in one direction?



- A) 12 B) 9 C) 6 D) 15 E) 24

15)

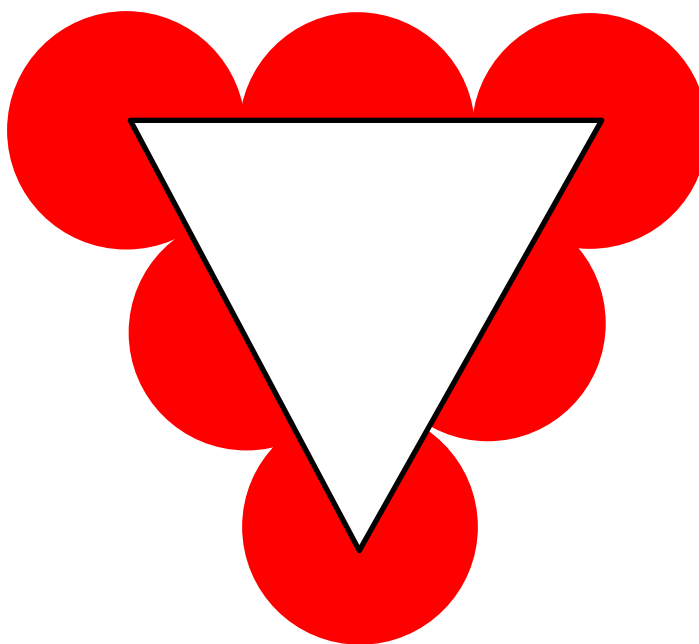


In the previous figure, triangle $\triangle CAB$ is an isosceles right triangle, and $\overline{CB}$ is the diameter of the gray circle. If the circumference of the white circle is twice the circumference of the gray circle and the area of $\triangle CAB$ is 81 units, what is the larger white circle's circumference?

- A) $3\pi\sqrt{2}$ B) 24π C) $6\pi\sqrt{2}$ D) $6\pi\sqrt{3}$ E) 36π

1	2	3	4	5	6	7	8	9	10
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16)



In the previous figure, an equilateral triangle has its vertices at the centers of circles. If each circle has the same area and the triangle area is $36\sqrt{3}$, what is the circumference of the outside of the figure?

- A) 24π B) 36π C) 2π D) 88π E) 220π

Solutions Chapter 7

Circles/ Arcs /Sectors

- | | | | |
|------|------|-------|-------|
| 1) B | 5) A | 9) A | 13) E |
| 2) C | 6) B | 10) E | 14) A |
| 3) E | 7) A | 11) C | 15) E |
| 4) C | 8) A | 12) B | 16) A |

1) One rotation around a circle = 360° . To find how many degrees eight and a half rotations are, multiply $360^\circ \cdot (8.5) = 3060$. Also note; (half rotation = $\frac{1}{2} \cdot 360^\circ = 180^\circ$, quarter rotation = $\frac{1}{4} \cdot 360^\circ = 90^\circ$)

3060 Answer B Difficulty Level 2

2) Since the circumference of the circle is given, find the radius, by working backward. The formula for the circumference of a circle is $C = 2\pi r$ or πd ; $30\pi = 2 \cdot \pi \cdot r$; solve for r ; $r = 15$. Since $p - 10 = r$, $p - 10 = 15$, solving for p , $p = 25$.

25 Answer C Difficulty Level 2.5

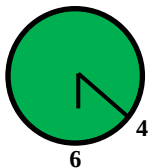
3) Since all circles total 360° around, the shaded (green) portion is $360 - 200 = 160^\circ$. This portion represents $\frac{160}{360}$ of the circle. Reduce the fraction. $\frac{4}{9}$ of the circle is shaded. (green)

4/9 Answer E Difficulty Level 5

4) Since the sum of the angles around a point always equals 360 degrees, $360/40$ would be the central angle of each slice = 9° . Therefore, the central angle of each slice is between 5 and 10 degrees.

Between 5° and 10° Answer C Difficulty Level 4.5

5) In an analog clock, if the minute hand is on the 4 and the hour hand is on the 6, the angle formed is shown in the following figure. Since the numbers on clocks are equally spaced from 1 through 12, each number represents $\frac{360^\circ}{12} = 30^\circ$. Therefore, the central angle formed is 60° ; $30^\circ(4 \text{ to } 5) + 30^\circ(5 \text{ to } 6) = 60^\circ$



60° Answer A Difficulty Level 5.5

6) To find the length of an arc, use the arc length formula (Arc length = $\frac{\text{central angle}^\circ}{360} \cdot 2\pi r$). Since A is the center of the circle, BC is the diameter, and the length of the radius is half the diameter ($\frac{1}{2} \cdot 12 = 6$); $r = 6$. Arc length DE = $\frac{60}{360} \cdot 2 \cdot \pi \cdot 6 = \frac{1}{6} \cdot 12\pi = 2\pi$. (Note: For the arc length formula, substitute πd for $2\pi r$ when using the circle's diameter).

2π Answer B Difficulty Level 7

7) To find the length of an arc, use the arc length formula. (Arc length = $\frac{\text{central angle}^\circ}{360} \cdot 2\pi r$). Since the length of the radius of the circle (AC) is given (6mm.), find the measure of the central angle ($\angle ACB$). Since all circles are 360° , the central angle is $360^\circ - 280^\circ = 80^\circ$. Finally, plug it in and simplify; $\frac{80}{360} \cdot 2\pi (6) = \frac{2}{9} \cdot 12\pi = \frac{8}{3} \pi$

$\frac{8}{3} \pi$ Answer A Difficulty Level 7.5

8) To find the area of a sector of a circle, use the formula (Sector Area = $\frac{\text{central angle}^\circ}{360} \cdot \pi r^2$). Since segment AO is from the center to the edge, AO is the radius of circle O. (Arc ACB is a semicircle 180°) Since arc AB is 45° , arc AC

must be; $180 - 45 = 135^\circ$. Therefore, the central angle ($\angle AOC$) is also 135° . Finally, plug it in and simplify. Sector Area = $\frac{135^\circ}{360} \cdot \pi (8)^2$ or 24π .

24π Answer A Difficulty Level 7.5

9) Since each circle has a circumference of 4π and each circle has the same radius of ($C=2\pi r$), $4\pi = 2\pi r$, so $r = 2$. That makes the length of $DE = 8$ and the length of $DF = 4$ (4 radii times 2 = 8, 2 radii times 2 = 4). Finally find the length of EF by using the Pythagorean Theorem where A =side DE , B =side DF , and C =side EF ($A^2 + B^2 = C^2$ / note*only used for right triangles). $8^2 + 4^2 = C^2$, $64 + 16 = C^2$, $C^2 = 80$, then take the square root of both sides of the equation and simplify. $C = \sqrt{80}$, $C = \sqrt{16 \cdot 5}$; $C = 4\sqrt{5}$

$4\sqrt{5}$ Answer A Difficulty Level 7.5

10) To find the length of an arc, use the arc length formula, $\frac{\text{central angle}^\circ}{360^\circ} \cdot 2\pi r$. Since $m \angle ROT$ is 120° , $m \angle RAT$ is $360^\circ - 120^\circ = 240^\circ$ and $\frac{240}{360} \cdot 2\pi (9\text{mm.}) = 12\pi \text{ mm.}$

12π Answer E Difficulty Level 7.5

11) If the area of rectangle $ABCD$ (length $\cdot$ width) is equal to 96 squared units and the length of $ABCD$ is given (12 units), the width is 8 units in length ($96 = 12 \cdot$ width). Therefore, the perimeter around the orange region is the length of segment BC (8 units) plus the length of the semicircle Arc BC with a radius of 4 units ($\frac{1}{2}$ the diameter (BC)). Now calculate and simplify the length of Arc BC . (Arc length = $\frac{\text{central angle}^\circ}{360} \cdot 2\pi r$). Arc $BC = \frac{180^\circ}{360^\circ} 2\pi (4) = \frac{1}{2} \cdot 8\pi = 4\pi$. The perimeter of the orange region is $8 + 4\pi$.

$8 + 4\pi$ Answer C Difficulty Level 8

12) To find the radius given the area of a sector and its central angle, use the sector area formula and work backward or solve for the missing variable. (Sector

Area = $\frac{\text{central angle}^\circ}{360} \cdot \pi r^2$). The central angle is not given but can easily be found by noticing that the other part of the circle covers 225° . The measure of angle MAX = $360^\circ - 225^\circ = 135^\circ$. Plug in the given area of the sector and the central angle to calculate the length of the radius, AX. Sector MAX Area $3\pi = \frac{135}{360} \cdot \pi r^2$. Next, solve for r by first dividing both sides of the equation by π ; $3 = \frac{135}{360} \cdot r^2$ then, reduce the fraction on the right-hand side of the equation to $\frac{3}{8}$; $3 = \frac{3}{8} \cdot r^2$ and, multiply both sides of the equation by the reciprocal $\frac{8}{3}$; $8 = r^2$. Finally, take the square root of both sides of the equation, $r = \sqrt{8} = 2\sqrt{2}$.

$2\sqrt{2}$ Answer B Difficulty Level 8

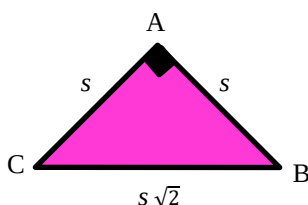
13) To find the area of the blue region, use the area of a sector formula (Sector Area = $\frac{\text{central angle}^\circ}{360} \cdot \pi r^2$). First, calculate the central angle of the blue region labeled $6x$. The sum of the three angles of the inside of the circle must sum to 360° . So, $4x + 5x + 6x = 360$; $15x = 360$, $x = 24$. Substitute back into the variable $6x$, $6(24) = 144^\circ$. Finally, find the area of the sector (blue region) Sector Area = $\frac{144}{360} \cdot \pi r^2$, reduce the fraction; $\frac{2}{5} \cdot \pi r^2$

$\frac{2}{5} \pi r^2$ Answer E Difficulty Level 8.5

14) When solving problems with circles, finding the length of the radius is usually the key. If the area of the gray tire is $\frac{81}{4\pi}$ Feet, (Area of a circle; $A = \pi r^2$), work backward to find the radius $\frac{81}{4\pi}$ feet = πr^2 , multiply both sides of the equation by π ; $\frac{81}{4} = \pi^2 r^2$, take the square root of both sides, $\pi r = \frac{9}{2}$, then multiply both sides of the equation by $\frac{1}{\pi}$ so $r = \frac{9}{2\pi}$. Since the gray tire's radius is three times the length of the pink tire's radius or the pink tire's radius is one-third the length of the gray tire's, the radius of the pink tire = $\frac{1}{3} \cdot \frac{9}{2\pi} = \frac{3}{2\pi}$. Finally, the circumference of the pink tire ($C = 2 \cdot \pi \cdot r$) $C = 2 \cdot \pi \cdot \frac{3}{2\pi} = 3$ feet. If the geometric contraption moves 36 feet, the pink tire rotates $\frac{36}{3} = 12$ times.

12 Answer A Difficulty Level 9.5

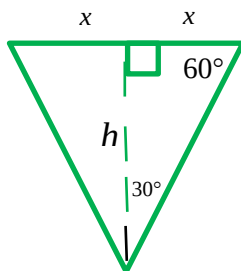
15) Triangle CAB is a right isosceles triangle which means it is a special right triangle (45° 45° 90°). Since A is the right (90°) angle, sides AB and AC must be congruent. Also given, triangle CAB has an area of 81, therefore
(Area of a triangle; $A = \frac{1}{2} \cdot b \cdot h$) $81 = \frac{1}{2} \cdot b \cdot h$; $bh = 162$; $b = h = \sqrt{162}$ or $9\sqrt{2}$.
Finally, calculate CB, which is the hypotenuse of triangle CAB and the diameter of the gray circle. Since each leg of the 45° - 45° - 90° triangle is $9\sqrt{2}$ units in length, CB must be $9\sqrt{2} \cdot \sqrt{2} = 9 \cdot 2 = 18$.



Since the diameter, CB is 18, the circumference of the gray circle is ($C = \pi d$); $C = 18\pi$ and the outside white circle has a circumference that is twice as large as the gray circle. $18\pi \cdot 2 = 36\pi$.

36\pi Answer E Difficulty Level 9

16) Since the triangle is equilateral, all the sides are congruent, and each interior angle is 60° . Draw the altitude and split the triangle into two 30° 60° 90° triangles. To find the height, label half the base as x , then multiply by $\sqrt{3}$, the height $h = x\sqrt{3}$. Since the area of a triangle $= \frac{1}{2} b \cdot h$, set $36\sqrt{3} = \frac{1}{2} b \cdot h$; the length of the base(b) is $2x$; substitute and solve for x , $36\sqrt{3} = \frac{1}{2} \cdot (2x) \cdot x\sqrt{3}$, $x^2\sqrt{3} = 36\sqrt{3}$, therefore $x = 6$. Since all sides of the triangle are congruent, they are all 12 units in length



Each side of the triangle is equal to 4 radii, where the radius of each circle = $\frac{1}{2}$ or 3 units. Find the perimeter of the figure, which consists of three semicircles and three circles covered partially by the corners of the triangle. The perimeter of the semicircles: $\frac{1}{2} \cdot (2\pi r)$ or $\frac{1}{2} \cdot (2\pi (3)) = 3\pi$. Since there are three semicircles, their total circumference is $3 \cdot 3\pi = 9\pi$. To find the perimeter of the remaining three partial circles, subtract 60 degrees from 360 = 300 degrees. The other three circles each have a perimeter showing of; (Arc length formula; $\frac{\text{central angle}^\circ}{360} \cdot 2\pi r$); $\frac{300}{360} \cdot 2\pi (3) = 5\pi$. There are three partial circles with a total perimeter showing of 15π . The perimeter of the figure is $9\pi + 15\pi = 24\pi$.

24π Answer A Difficulty Level 10